Update Rule: How we make weight from the old weight + ↑. of the Loss w.r.t. the old weight.

$\frac{dL}{dw_{ij}}$ = What we computed during backprop.

↳ "Blame" → what part of the Loss we attribute to this particular weight

Update Rule:

New weight = old weight − learning rate * derivative of Loss w.r.t. old weight

For some weight w_{ij}

old weight = $w_{ij(t-1)}$

new weight = w_{ij+}

$$w_{ij+} = w_{ij(t-1)} - \rho \frac{dL}{dw_{ij(t-1)}}$$

(ρ: learning rate)

$\frac{dL}{dw_{ij(t-1)}}$ → "Blame" — Part of error weight is responsible for (direction it should move)

ρ → Step size in the right direction we want to take.

Update Rule → Changes the weight to a step towards better parameters scaled by responsibility for error

Really quick example:

Old weight = 10

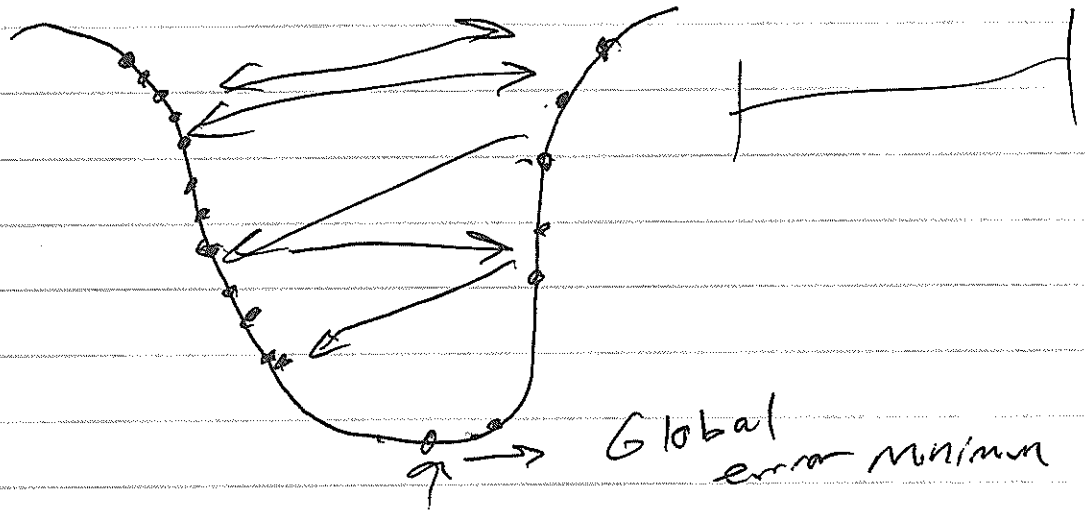
$\eta = 0.001$

$\frac{dL}{dw} = 0.3$

Has a huge impact on how fast your network trains.

New weight = $(0 - (.001)(.3)$

New weight = 9.997



Regularization:

L1 Regularization Error Function

$$\text{Error} = \text{Error}(y, t) + \lambda \sum_{i=1}^{N} |w_i|$$

(y: output, t: target, λ: parameter)

L2 Regularization

$$\text{Error} = \text{Error}(y, t) + \lambda \sum_{i=1}^{N} w_i^2$$